CANDIDATE NAME (CG) ________________________ *INDEX NUMBER* _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2008

BIOLOGY
Higher 2

Paper 2 (Core) - 9747

Saturday
23 August 2008

2 hours

Additional materials: Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

At the end of examination,

1. fasten all your work securely together;
2. enter the numbers of the Section B and C questions you have answered in the grid opposite.

FOR EXAMINER'S USE	
Section B	
1	**/10**
2	**/10**
3	**/10**
4	**/10**
5	**/10**
6	**/10**
7	**/10**
8	**/10**
Section B	
	/20
TOTAL	**/100**

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [ ] at the end of each question or part question.

This question paper consists of 21 printed pages.

Section A (80 marks)

Answer all questions in the space provided

Question 1

Figure 1.1 below shows a variety of cells dividing by mitosis.

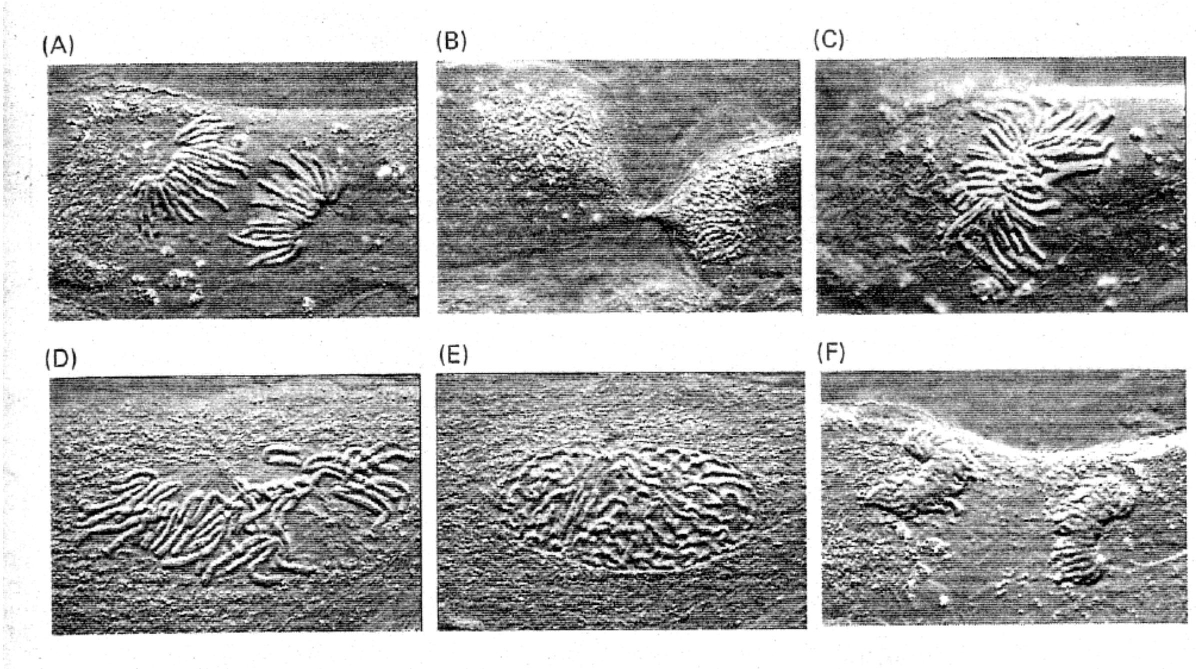

Figure 1.1

(a) Using the letters assigned to each cell (ie. A, B, ….), arrange the cells in the correct sequence that they will undergo during division. [1]

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(b) Identify the mitotic stages the following cells are in with reasons. [6]

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F - ……………………………………………………………………………………………

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(c) Discuss the significance of mitotic division to a living organism. [3]

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Question 2

Figure 2.1 illustrates the structure of a typical synapse.

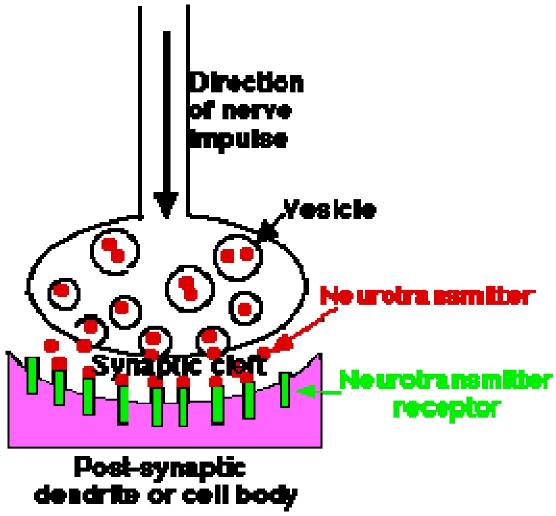

Figure 2.1

(a) Provide alternative names for the following: [2]

Vesicle - ____________________________

Neurotransmitter - _____________________

(b) Describe how an electrical nervous impulse arriving at the pre-synaptic knob could be converted into a chemical form in the synaptic cleft. [3]

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(c) Describe how a nervous impulse is generated on the membrane of the post synaptic dendrite. [2]

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(d) Chemical X, which has a similar shape to the neurotransmitter, is introduced into the synapse. Explain the effect that it would have on nervous transmission across the synapse. [2]

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(e) Suggest why synapses are unable to function at low temperatures. [1]

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[Total: 10 marks]

Question 3

Figure 3.1 is an illustration of a phylogenetic tree.

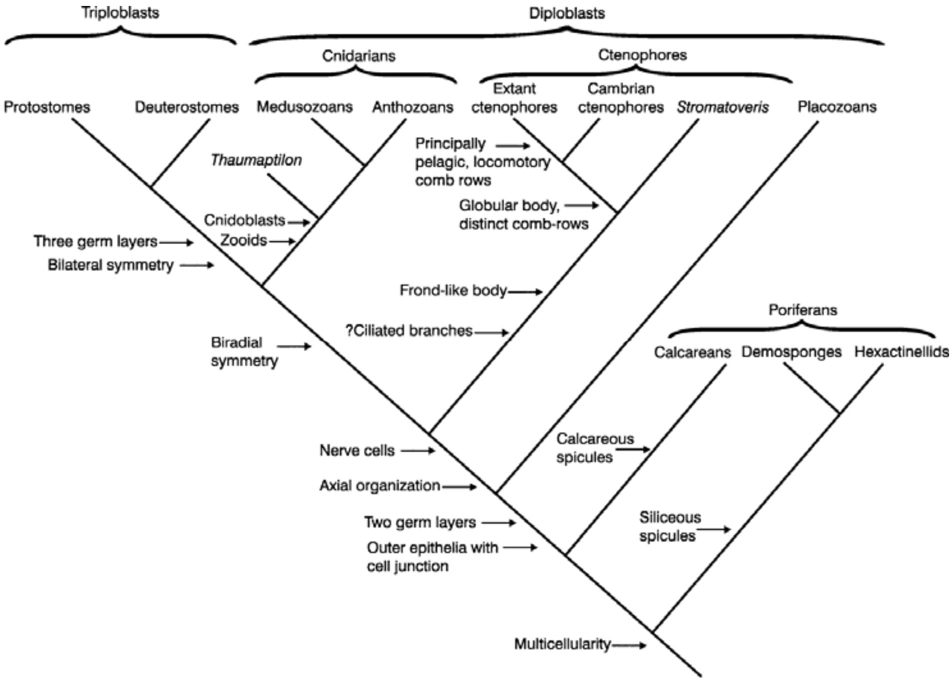

Figure 3.1

(a) Give the general name for such a tree and explain its usefulness. [1]

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(b) Using information from Figure 3.1, suggest and explain if anthozoans are more closely related to medusozoans or protostomes. [2]

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Figure 3.2 shows the development of finches on the Galapagos Islands.

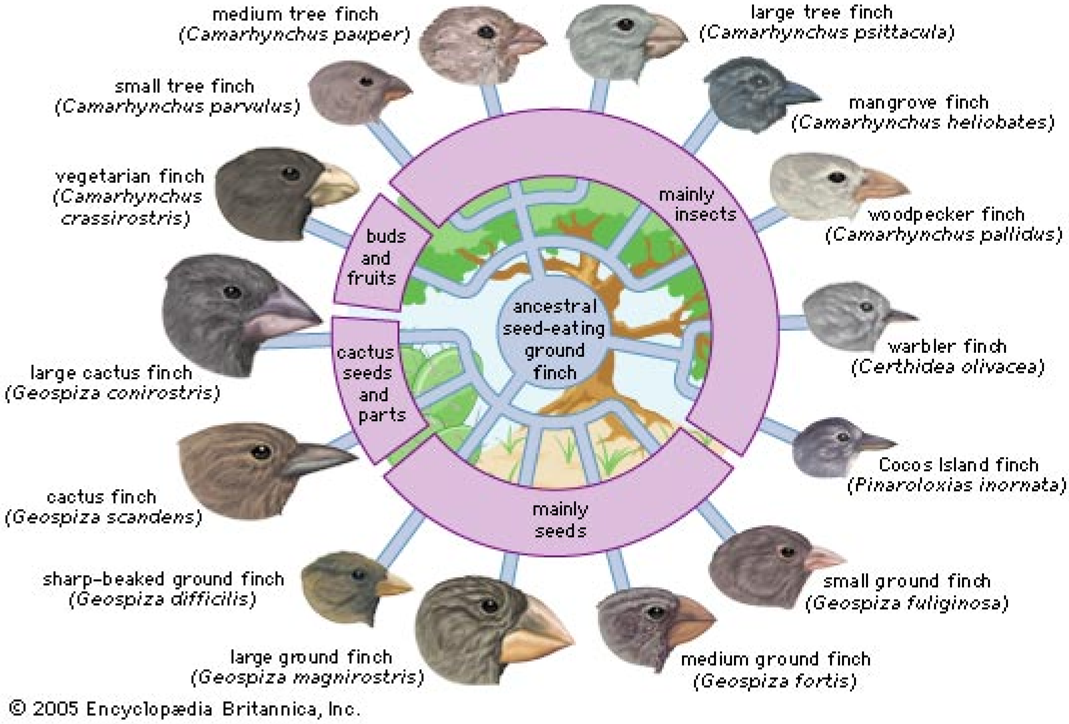

Figure 3.2

(c) (i) Discuss how the ancestral seed-eating ground finch speciated into a variety of different species. [4]

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(ii) Compare the type of speciation you just described with sympatric speciation. [2]

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(d) According to the evolutionary concept of natural selection, recessive disease alleles like those causing sickle cell anaemia and cystic fibrosis should be selected against. This means that the frequency of these alleles in the human population should be low. However, this is not the case. The frequency of these alleles remain relatively high. Suggest a reason for this. [1]

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[Total: 10 marks]

Question 4

(a) A diploid plant species is capable of both self- and cross-pollination. Two pure bred varieties **A** and **B** of this plant species were identified. Individuals of variety **A** have heights ranging from 21 to 25 cm. They all have round leaves and are resistant to the herbicide glyphosate. Individuals of variety **B** have heights ranging from 56 to 60 cm. They all have wrinkled leaves and are sensitive to the herbicide glyphosate.

When **A** was crossed with **B**, all F1 progeny had round leaves and was glyphosate resistant. Their height distribution is shown in the table below.

Height / cm	Number of plants
26 – 30	1
31 – 35	10
36 – 40	54
41 – 45	51
46 – 50	8
51 – 55	3

Table 4.1

When the F1 plants were self-pollinated, the F2 progeny had the following phenotypes:

Height / cm	Round leaf & resistant to glyphosate	Wrinkled leaf & resistant to glyphosate	Round leaf & sensitive to glyphosate	Wrinkled leaf & sensitive to glyphosate
21 – 25	15	1	0	5
26 – 30	88	2	2	28
31 – 35	114	4	3	35
36 – 40	152	5	5	48
41 – 45	152	5	6	47
46 – 50	116	4	4	36
51 – 55	86	2	3	27
56 – 60	13	0	0	4
Sub-total	736	23	23	230

Table 4.2

With reference to **Tables 4.1** and **4.2**, answer the following questions

(i) Are the alleles responsible for wrinkled leaves and glyphosate resistance dominant or recessive? Briefly explain your answer. [2]

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(ii) How many genes and alleles control the glyphosate response? [1]

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(iii) With reference to **Tables 4.1** and **4.2**, explain your answer. [2]

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(iv) Given that the critical value $\chi^2_{(0.05,\ 3)}$ is 7.815, conduct a chi square analysis to determine if the inheritance of leaf shape and glyphosate response follows Mendel's law of independent assortment. Refer to Annex A for the formula.[5]

Question 5

2,6-dichlorophenolindophenol (DCPIP) is a blue dye that can be converted to colourless reduced DCPIP by gaining electrons. This is summarized below.

$$\underset{\text{(blue)}}{\text{DCPIP}} \xrightarrow[\textit{electrons}]{} \underset{\text{(colourless)}}{\text{Reduced DCPIP}}$$

A chloroplast suspension was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were prepared and treated in different ways. The colour of the tube contents was recorded at the start and after 15 minutes. This information is summarized in the table.

			Colour	
Tube	**Contents**	**Treatment**	**At start**	**After 15 minutes**
A	2cm^3 chloroplast suspension 6cm^3 DCPIP	Tube kept in bright light	Blue/ green	Green
B	2cm^3 chloroplast suspension 6cm^3 DCPIP	Tube kept in dark	Blue/ green	Blue/ green
C	2cm^3 buffer solution 6cm^3 DCPIP	Tube kept in bright light	Blue	Blue

(a) (i) Tube C was included as a control. Explain why this control was necessary in the investigation. [1]

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(ii) Explain the colour of tube A after 15 minutes. [2]

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(b) Describe how reduced NADP is involved in the light-independent reactions of photosynthesis. [2]

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(c) Apart from the difference in the presence or absence of oxygen, list a difference between aerobic and anaerobic respiration. [1]

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(d) During glycolysis, NAD is reduced. Explain what happens to this reduced NAD when the cell is respiring anaerobically. [2]

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(e) Explain why, in the absence of oxygen, the electron transfer chain in the mitochondria does not operate. [2]

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[Total: 10 marks]

Question 6

The following figure 6.1 is a simplified diagrammatic representation of a cell signalling pathway.

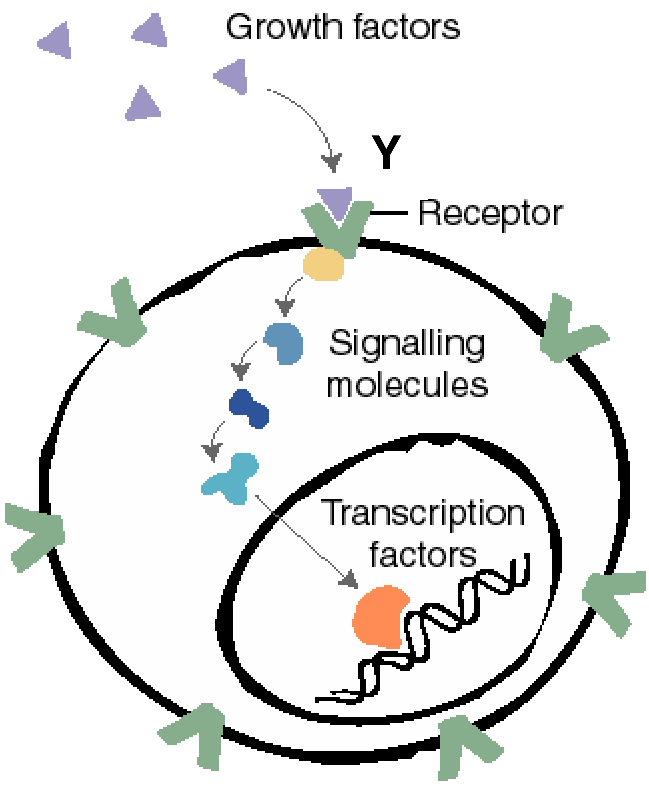

Figure 6.1

(a) With the use of a specific example, discuss the role of structure Y. [3]

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(b) Suggest why transcription factors are the normal end products of such a pathway. [1]

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Figure 6.2 shows a simplified representation of the downstream cellular effects of the dimerization and activation of the VEGF/VEGFR-2 complex in an endothelial cell.

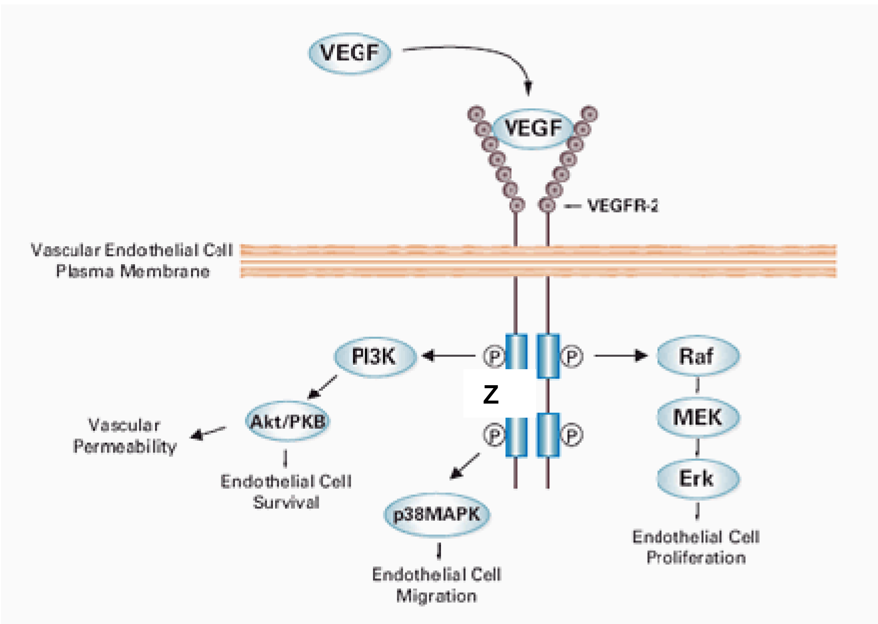

Figure 6.2

(c) Label clearly on the diagram the three main stages of cell signalling. [1]

(d) Suggest a role for the phosphorylated molecules labelled Z. [2]

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(e) Using information from Figure 6.2 explain why this signal transduction pathway may lead to cancerous expression of the endothelial cell. [3]

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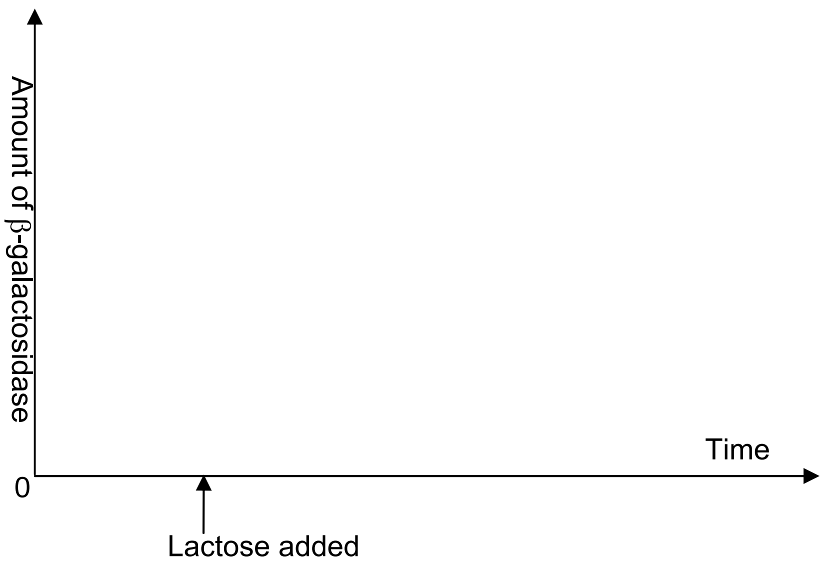
[Total: 10 marks]

Question 7

(a) A study of the bacterium's chromosome showed that it possessed a *lac* operon to allow the bacterium to metabolise lactose.

(i) On the axes below, draw a graph showing how the amount of *β*-galactosidase in the bacterium is affected by the addition of lactose. [1]



(ii) Describe how the following pathway is turned on in *E.coli*. [4]

Lactose outside the cell $\xrightarrow{\text{permease}}$ lactose inside the cell

lactose inside the cell $\xrightarrow{\beta\text{-galactosidase}}$ glucose + galactose

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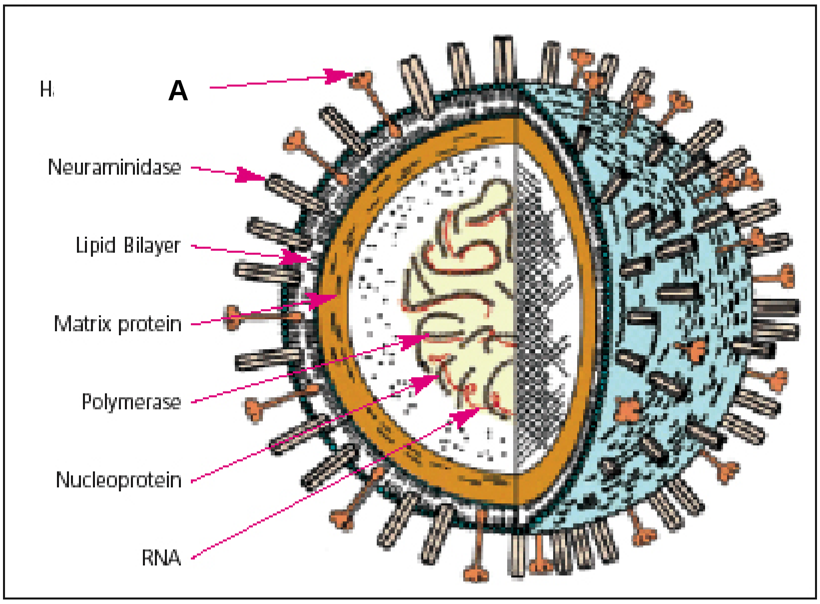
(b) Fig. 7.1 shows the main structural features of an influenza virion.



Source: Chotani RA (2006) *The impact of pandemic influenza on public health.*[16]

Fig. 7.1

(i) Identify molecule **A** and describe its role in infection. [2]

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(ii) Describe the role of neuraminidase in the life cycle of the influenza virus. [1]

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(c) Table 7.1 shows an excerpt of the confirmed human cases of avian influenza reported by the World Health Organisation (WHO).

Table 7.1: Cumulative Number of Confirmed Human Cases of Avian Influenza /(H5N1)
reported to WHO 19 June 2008

Country	2003		2004		2005		2006		2007		2008		
	cases	deaths	cases	deaths	cases	deaths	cases	deaths	cases	deaths	cases	deaths	case
Azerbaijan	0	0	0	0	0	0	8	5	0	0	0	0	8
Bangladesh	0	0	0	0	0	0	0	0	0	0	1	0	1
Cambodia	0	0	0	0	4	4	2	2	1	1	0	0	7
China	1	1	0	0	8	5	13	8	5	3	3	3	30
Djibouti	0	0	0	0	0	0	1	0	0	0	0	0	1
Egypt	0	0	0	0	0	0	18	10	25	9	7	3	50
Indonesia	0	0	0	0	20	13	55	45	42	37	18	15	135
Iraq	0	0	0	0	0	0	3	2	0	0	0	0	3

(i) Avian flu is typically described as “influenza caused by viruses adapted to birds”. Explain the occurrence of human cases of avian influenza.[2]

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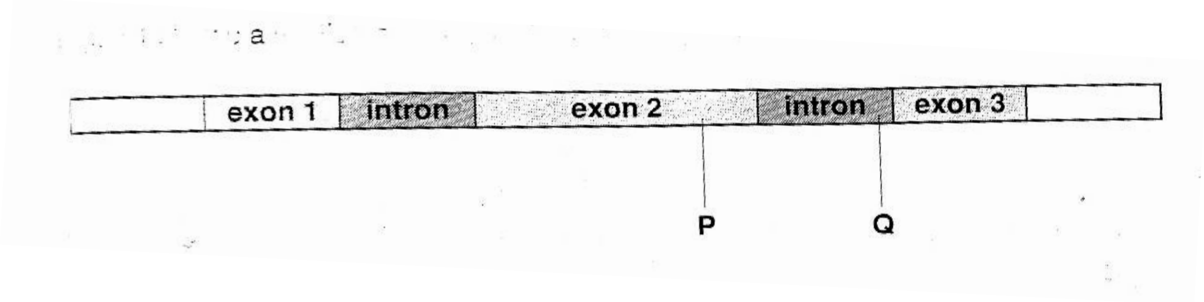
Question 8

Figure 8.1 shows a section of DNA, containing one gene.



(a) (i) State one way in which the structure of this gene indicates that it is from a eukaryote. [1]

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(ii) Using the same diagrammatic format as Fig 8.1, show what the pre-mRNA and mature mRNA will be for this gene.

Pre-mRNA [1]

Mature mRNA [1]

(iii) Describe the process that has occurred to when a pre-mRNA is converted to a mature mRNA. [3]

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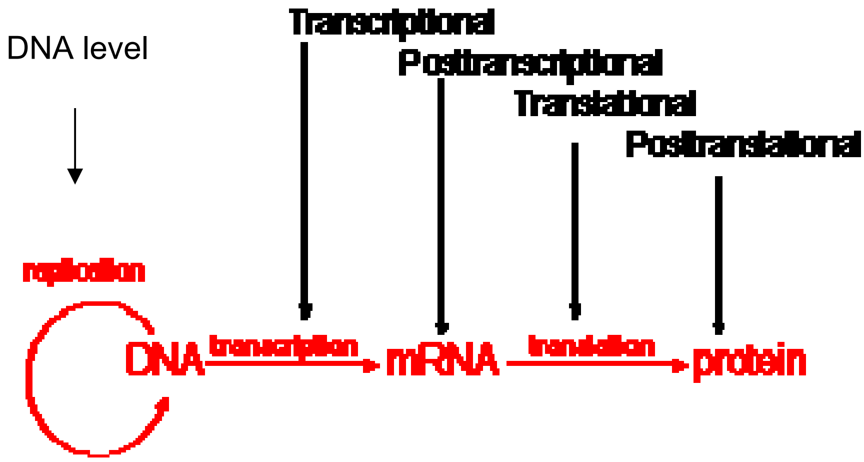
Control of gene expression can occur at any step: at the DNA level, transcriptional level, post-transcriptional, translational level and post-translational level.



(iv) State two ways to control the expression of genes at the DNA level and explain how these modifications work to control gene expression. [4]

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[Total: 10 marks]

Section B (20 marks)

Answer **only one** question.

Write your answer on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Question 9

(a) Briefly explain why membranes are described as fluid mosaic in structure. [6]

(b) Describe how the membrane affects the entry and exit of substances in and out of a cell. [6]

(c) Using appropriate examples, describe how substances can be brought across a membrane by facilitated diffusion, active transport and endo/exo-cytosis. [8]

Question 10

(a) Describe the structure of DNA & its arrangement in a chromosome. [8]

(b) Distinguish between DNA replication and transcription, indicating the important features of each process. [6]

(c) Explain, with an example, how a change in the sequence of the DNA nucleotide may affect the amino acid sequence in a protein and hence the phenotype of the organism. [6]

END OF PAPER

Annex A

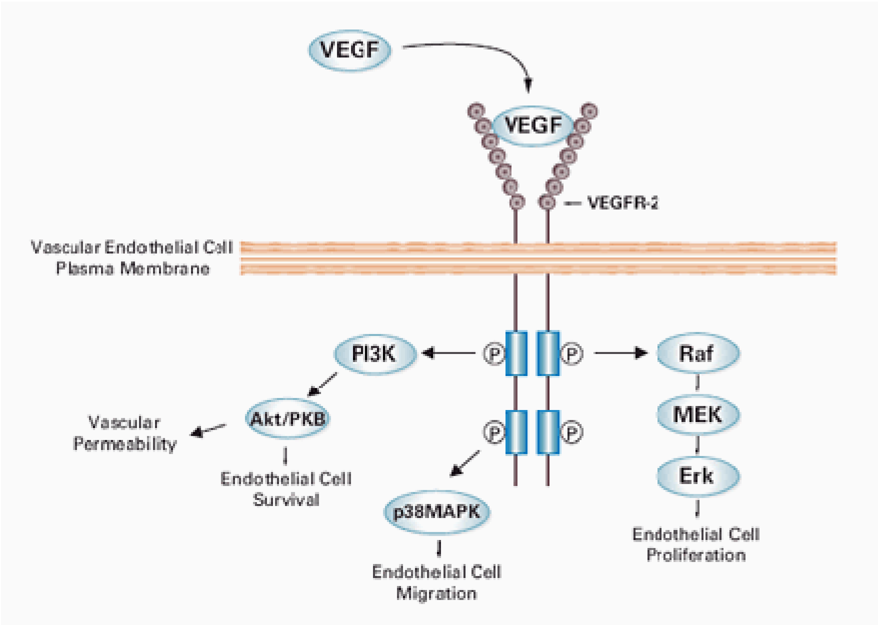

Figure 6.2

χ^2 test $\qquad \chi^2 = \Sigma \frac{(O-E)^2}{E} \qquad v = c - 1$

Key to symbols

*s = standard deviation	n = sample size (number of observations)	E = expected 'value'
Σ = 'sum of '	v = degrees of freedom	
x = observation	c = number of classes	
$\bar{x}$ = mean	O = observed 'value'	